PRKACA fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes PRKACA gene fusions from the msk_impact_50k_2026 study, covering 35 fusion events. 24/25 fusions with known frame status (96.0%) were in-frame. 29/30 fusions (96.7%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test (p=0.214286, not statistically significant at alpha=0.05). No literature benchmark is configured for PRKACA.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 35 analyzed fusion events, identifying 2 distinct fusion partner genes. The most frequent partner was DNAJB1, observed in 34 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 24/24 (100.0%) of in-frame fusions retained the domain, p=0.214286 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 9.9999e-06.

Domain disruption

Domain-disruption analysis was skipped: PRKACA has no disruption_required_domains configured; domain-disruption analysis was skipped.

Junction concentration

100% of 27 fusions with this gene as the 3' partner join at exon 2, against 39% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=9.9999e-06, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 28 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 16 aa (exon 1/2 boundary; label-permutation p=0.038961). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -29 aa from the nearest configured domain boundary at 45 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=27) and "not_retained" (n=1). For Frame_status, retained: 22/22 (100.0%, 95% CI 85.1%-100.0%); not_retained: 0/1

(0.0%, 95% CI 0.0%-79.3%). Welch's t-test comparing Tumor_variant_count between groups gave means of 26.7037 vs 61, p=unavailable.

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for PRKACA in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (11 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 1 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: PRKACA has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: PRKACA has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection reported: no candidate window of the configured widths (25, 50, 100, 200) met the minimum-event-count guard (min_events_per_window=4) on both sides of the split

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
DNAJB1	34	0.9714285714285714	0.5000004342923104	None	1.0	0.5289472198843083	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.7741542524875997	1
PIK3R2	1	0.02857142857142857	0.5000004342923104	None	0.0	0.5289472198843083	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.23308282391617127	2

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
16	27	0	0	1	None	0.03571428571428571	1.4471580313422192

domain_retention tables

frame_domain_contingency_table

22	5
0	1

frame_domain_contingency_tables_by_tumor_type

stratum	table
FLC	[18, 4], [0, 0]
HCC	[2, 1], [0, 0]
HGSOC	[0, 0], [0, 1]
IPMN	[1, 0], [0, 0]
PAAD	[1, 0], [0, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	28	27	1	0	0.8307086614173228

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
3	28	33	0.8484848484848485

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
DNAJB1	34	34	34	30
PIK3R2	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
19	14227000	14228000	5	EVT-P-0003615-T01-IM5-17, EVT-P-0010266-T01-IM5-30, EVT-P-0035557-T01-IM6-25, EVT-P-0047568-T01-IM6-7, EVT-P-0047946-T01-IM6-9
19	14222000	14223000	4	EVT-P-0006307-T01-IM5-26, EVT-P-0017715-T02-IM6-14, EVT-P-0053860-T01-IM6-10, EVT-P-0059665-T01-IM7-1
19	14218000	14219000	3	EVT-P-0004881-T01-IM5-13, EVT-P-0013377-T01-IM5-2, EVT-P-0014491-T01-IM6-32
19	14220000	14221000	3	EVT-P-0005135-T01-IM5-31, EVT-P-0005797-T01-IM5-20, EVT-P-0006454-T01-IM5-27
19	14224000	14225000	3	EVT-P-0010050-T01-IM5-29, EVT-P-0037291-T01-IM6-4, EVT-P-0038951-T01-IM6-6
19	14219000	14220000	2	EVT-P-0017490-T01-IM6-34, EVT-P-0037861-T01-IM6-5

chromosome	bin_start	bin_end	n_events	event_ids
19	14225000	14226000	2	EVT-P-0021668-T01-IM6-19, EVT-P-0023643-T01-IM6-21
19	14226000	14227000	2	EVT-P-0016186-T01-IM6-33, EVT-P-0028426-T01-IM6-23
19	14228000	14229000	2	EVT-P-0009917-T01-IM5-28, EVT-P-0055056-T01-IM6-11
19	14205000	14206000	1	EVT-P-0050640-T01-IM6-35
19	14221000	14222000	1	EVT-P-0025933-T01-IM6-22

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
16	27	27	9981	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
three_prime	2	27	27	1.0	0.3896841539701424	11.050755855369616
five_prime	8	1	1	1.0	0.14165099589361785	0.8478803677755075

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059665-T01-IM7-1	P-0059665-T01-IM7	P-0059665	Hepatobiliary Cancer	HCC	DNAJB1-PRKACA	DNAJB1	in-frame	three_prime	14222812	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion: c.212-173:DNAJB1_c.47-4591:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA
EVT-P-0013377-T01-IM5-2	P-0013377-T01-IM5	P-0013377	Hepatobiliary Cancer	FLC	DNAJB1-PRKACA	DNAJB1	in-frame	three_prime	14218681	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion: c.212-162:DNAJB1_c.47-460:PRKACAdel | Protein fusion: in frame (DNAJB1-PRKACA)	NA
EVT-P-0021668-T02-IM7-3	P-0021668-T02-IM7	P-0021668	Hepatobiliary Cancer	HCC	DNAJB1-PRKACA	DNAJB1	in-frame	three_prime	14225506	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion: c.212-423:DNAJB1_c.46+2808:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0037291-T01-IM6-4	P-0037291-T01-IM6	P-0037291	Pancreatic Cancer	PAAD	DNAJB1-PRKACA	DNAJB1	in-frame	three_prime	142245444	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion (DNAJB1 exon 1 fused to PRKACA exons 2-10): c.212-523:DNAJB1_c.46+3770:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA
EVT-P-0037861-T01-IM6-5	P-0037861-T01-IM6	P-0037861	Hepatobiliary Cancer	FLC	DNAJB1-PRKACA	DNAJB1	unknown	three_prime	142196318	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion: c.517:DNAJB1_c.47-1417:PRKACAdel | Protein Fusion: mid-exon {DNAJB1:PRKACA}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038951-T01-IM6-6	P-0038951-T01-IM6	P-0038951	Pancreatic Cancer	IPMN	DNAJB1-PRKACA	DNAJB1	in-frame	three_prime	14224096	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion: c.212-7:DNAJB1_c.46+4218:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA
EVT-P-0047568-T01-IM6-7	P-0047568-T01-IM6	P-0047568	Hepatobiliary Cancer	FLC	DNAJB1-PRKACA	DNAJB1	in-frame	three_prime	14227968	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) rearrangement: c.212-269:DNAJB1_c.46+346:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA
EVT-P-0047568-T02-IM6-8	P-0047568-T02-IM6	P-0047568	Hepatobiliary Cancer	FLC	DNAJB1-PRKACA	DNAJB1	in-frame	three_prime	14227968	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) rearrangement: c.212-269:DNAJB1_c.46+346:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0047946-T01-IM6-9	P-0047946-T01-IM6	P-0047946	Hepatobiliary Cancer	FLC	DNAJB1-PRKACA	DNAJB1	in-frame	three_prime	14227017	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion: c.211+437:DNAJB1_c.46+1297:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA
EVT-P-0053860-T01-IM6-10	P-0053860-T01-IM6	P-0053860	Hepatobiliary Cancer	FLC	DNAJB1-PRKACA	DNAJB1	in-frame	three_prime	14222854	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) rearrangement: c.212-174:DNAJB1_c.47-4633:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA
EVT-P-0055056-T01-IM6-11	P-0055056-T01-IM6	P-0055056	Hepatobiliary Cancer	FLC	DNAJB1-PRKACA	DNAJB1	in-frame	three_prime	14228258	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion: c.212-526:DNAJB1_c.46+56:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004881-T01-IM5-13	P-0004881-T01-IM5	P-0004881	Hepatobiliary Cancer	FLC	PRKACA-DNAJB1	DNAJB1	in-frame	three_prime	142187914	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion (DNAJB1 exon 1 fused with PRKACA exons 2-10) : c.212-414:DNAJB1_c.47-573:PRKACAdel (PMID: 24578576) | Protein fusion: in frame (DNAJB1-PRKACA)	NA
EVT-P-0017715-T02-IM6-14	P-0017715-T02-IM6	P-0017715	Hepatobiliary Cancer	FLC	PRKACA-DNAJB1	DNAJB1	in-frame	three_prime	142221213	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion (DNAJB1 exon 1 fused in-frame with PRKACA exons 2-10): NM_006145.1:c.212-373:DNAJB1_c.47-3902:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003615-T01-IM5-17	P-0003615-T01-IM5	P-0003615	Hepatobiliary Cancer	FLC	PRKACA-DNAJB1	DNAJB1	in-frame	three_prime	14227077	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			c.212-276:DNAJB1_c.46+1237:PRKACAdel | Protein fusion: in frame (DNAJB1-PRKACA)	NA
EVT-P-0021668-T01-IM6-19	P-0021668-T01-IM6	P-0021668	Hepatobiliary Cancer	FLC	PRKACA-DNAJB1	DNAJB1	in-frame	three_prime	14225506	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion (DNAJB1 exon 1 fused with PRKACA exons 1-10 which includes the kinase domain.) : c.212-423:DNAJB1_c.46+2808:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005797-T01-IM5-20	P-0005797-T01-IM5	P-000597	Hepatobiliary Cancer	HCC	PRKACA-DNAJB1	DNAJB1	in-frame	three_prime	142202529	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion (DNAJB1 exon 1 fused to PRKACA exons 2-10): c.212-213:DNAJB1_c.47-2038:PRKACAdel | Protein fusion: in frame (DNAJB1-PRKACA)	NA
EVT-P-0023643-T01-IM6-21	P-0023643-T01-IM6	P-002343	Hepatobiliary Cancer	FLC	PRKACA-DNAJB1	DNAJB1	in-frame	three_prime	142256312	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion (DNAJB1 exon 1 fused with PRKACA exons 2-10) : c.212-467:DNAJB1_c.46+2682:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025933-T01-IM6-22	P-0025933-T01-IM6	P-0025933	Hepatobiliary Cancer	FLC	PRKACA-DNAJB1	DNAJB1	in-frame	three_prime	14221557	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion (DNAJB1 exon 1 fused with PRKACA exons 2-10) : c.212-136:DNAJB1_c.47-3336:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA
EVT-P-0028426-T01-IM6-23	P-0028426-T01-IM6	P-0028426	Hepatobiliary Cancer	FLC	PRKACA-DNAJB1	DNAJB1	in-frame	three_prime	14226460	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion (DNAJB1 exon 1 fused with PRKACA exons 2-10) : c.212-467:DNAJB1_c.46+2682:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035557-T01-IM6-25	P-0035557-T01-IM6	P-0035557	Hepatobiliary Cancer	FLC	PRKACA-DNAJB1	DNAJB1	in-frame	three_prime	142273229	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion (DNAJB1 exon 1 fused with PRKACA exons 2-10) : c.212-112:DNAJB1_c.46+985:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA
EVT-P-0006307-T01-IM5-26	P-0006307-T01-IM5	P-0006307	Hepatobiliary Cancer	FLC	PRKACA-DNAJB1	DNAJB1	unknown	three_prime	142225229	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA: (NM_002730) Fusion (DNAJB1 exons 1 to 2 fused with PRKACA exons 2 to 10) : c.396:DNAJB1_c.47-4308:PRKACA | Protein fusion: mid-exon (DNAJB1-PRKACA)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006454-T01-IM5-27	P-0006454-T01-IM5	P-0006454	Hepatobiliary Cancer	FLC	PRKACA-DNAJB1	DNAJB1	unknown	three_prime	14220992	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			PRKACA (NM_002730) - DNAJB1 (NM_006145) rearrangement : c.47-2771:PRKACA_c.517:DNAJB1del | Protein fusion: mid-exon (DNAJB1-PRKACA)	NA
EVT-P-0009917-T01-IM5-28	P-0009917-T01-IM5	P-0009917	Hepatobiliary Cancer	FLC	PRKACA-DNAJB1	DNAJB1	in-frame	three_prime	14228299	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion (DNAJB1 exon 1 fused in frame with PRKACA exons 2-10): c.211+490:DNAJB1_c.46+15:PRKACAdel | Protein fusion: in frame (DNAJB1-PRKACA)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010050-T01-IM5-29	P-0010050-T01-IM5	P-0010050	Hepatobiliary Cancer	FLC	PRKACA-DNAJB1	DNAJB1	in-frame	three_prime	14224485	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) Fusion (DNAJB1 exon 1 fused to PRKACA exons 2-10):c.212-35:DNAJB1_c.46+3829:PRAKACA1del | Protein fusion: in frame (DNAJB1-PRKACA)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010266-T01-IM5-30	P-0010266-T01-IM5	P-0010266	Hepatobiliary Cancer	FLC	PRKACA-DNAJB1	DNAJB1	unknown	three_prime	14227277	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion (mid-exon fusion of DNAJB1 exons 1-2 with PRKACA exons 2-10; one of the breakpoints is within DNAJB1 exon 2): c.463:DNAJB1_c.46+1037:PRKACAdel | Protein fusion: mid-exon (DNAJB1-PRKACA)	NA

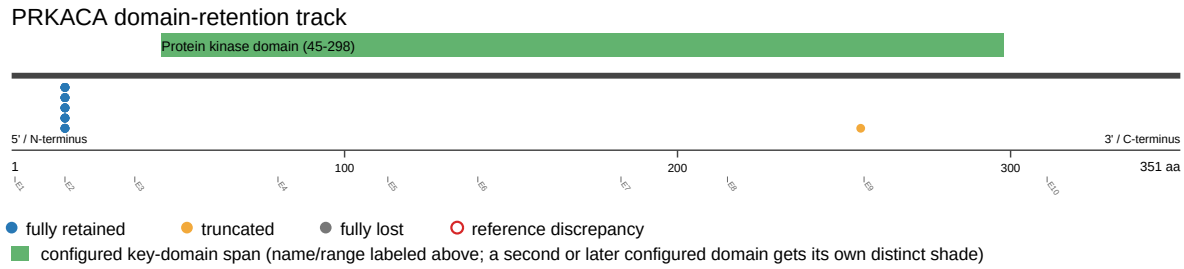
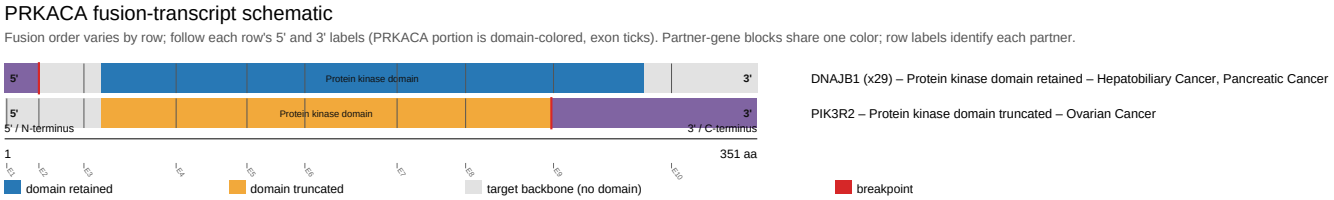
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005135-T01-IM5-31	P-0005135-T01-IM5	P-0005135	Hepatobiliary Cancer	FLC	PRKACA-DNAJB1	DNAJB1	in-frame	three_prime	14220905	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion (DNAJB1 exon 1 fuses with PRKACA exons 2-10): c.212-302:DNAJB1_c.47-2684:PRKACAdel | Protein fusion: in frame (DNAJB1-PRKACA)	NA
EVT-P-0014491-T01-IM6-32	P-0014491-T01-IM6	P-0014491	Hepatobiliary Cancer	HCC	PRKACA-DNAJB1	DNAJB1	unknown	three_prime	14218318	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			PRKACA (NM_002730)_DNAJB1 (NM_006145) rearrangement: c.47-97:PRKACA_c.491:DNAJB1del | Protein fusion: mid-exon (DNAJB1-PRKACA)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016186-T01-IM6-33	P-0016186-T01-IM6	P-0016186	Hepatobiliary Cancer	FLC	PRKACA-DNAJB1	DNAJB1	in-frame	three_prime	142260003	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion (DNAJB1 exon 1 fused with PRKACA exons 2-10) : c.211+346:DNAJB1_c.46+2311:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA
EVT-P-0017490-T01-IM6-34	P-0017490-T01-IM6	P-0017490	Hepatobiliary Cancer	FLC	PRKACA-DNAJB1	DNAJB1	in-frame	three_prime	142199962	2	16	True	16	True	exact match	retained	1.0	False	Protein kinase domain			DNAJB1 (NM_006145) - PRKACA (NM_002730) fusion (DNAJB1 exon 1 fused to PRKACA exon 2): c.212-129:DNAJB1_c.47-1741:PRKACAdel | Protein Fusion: in frame {DNAJB1:PRKACA}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0050640-T01-IM6-35	P-0050640-T01-IM6	P-0050640	Ovarian Cancer	HGSOC	PRKACA-PIK3R2	PIK3R2	out-of-frame	five_prime	142055555	8	255	True	255	True	exact match	disrupted	0.830708661417 3228	True			Protein kinase domain	PRKACA (NM_002730) - PIK3R2 (NM_005027) rearrangement: c.766-951:PRKACA_c.816-34:PIK3R2inv | Protein Fusion: out of frame {PRKACA:PIK3R2}	NA

Figures

domain retention outliers



fusion schematic